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Authors

Mr. Lucas Buffan
Corresponding Author
Submitting Author

ORCiD
https://orcid.org/0000-0002-2353-1432

Affiliations

- Institut des Sciences de l'Evolution de Montpellier, Université de Montpellier, CNRS, IRD, Place Eugène Bataillon, 34095 Montpellier cedex 5, France

Dr. Fabien L. Condamine

ORCiD
https://orcid.org/0000-0003-1673-9910

Affiliations

- Institut des Sciences de l'Evolution de Montpellier, Université de Montpellier, CNRS, IRD, Place Eugène Bataillon, 34095 Montpellier cedex 5, France

Dr. Sara Varela

Affiliations

- Centro de Investigación Mariña, Departamento de Ecoloxía e Bioloxía Animal, Universidade de Vigo, Campus Lagoas-Marcosende, 36310 Vigo, Spain

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Palaeoclimate alone fails to explain the asymmetry of the Great American Biotic Interchange

Lucas Buffan[1]*, Fabien L. Condamine[1], Sara Varela[2]

[1] Institut des Sciences de l'Evolution de Montpellier, Université de Montpellier, CNRS, IRD, Place Eugène Bataillon, 34095 Montpellier cedex 5, France.

[2] Centro de Investigación Mariña, Departamento de Ecoloxía e Bioloxía Animal, Universidade de Vigo, Campus Lagoas-Marcosende, 36310 Vigo, Spain

* Corresponding author: lucas.l.buffan@gmail.com

Statement of authorship

S.V. and L.B. designed the project. L.B. ran the simulations, performed the analyses, produced the figures and wrote the manuscript with inputs from all co-authors.

Data availability statement

R scripts for generating, analysing and visualising the data are available at https://github.com/Buffan3369/GABI-gen3sis.git. Simulation landscapes and config files are available at https://doi.org/10.6084/m9.figshare.31160704.

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Abstract

Asymmetric colonisation patterns often arise from large-scale biotic exchanges involving two source areas. *ca*. 5 million years ago, the closure of the Panamanian seaway facilitated the admixture between North and South American faunas, an event termed Great American Biotic Interchange (GABI). The ecological mechanisms underlying the asymmetry of the GABI, incontestably in favour of North American lineages, remain debated. Here, coupling a mechanistic eco-evolutionary simulator with palaeoclimatic reconstructions, we test fundamental hypotheses relating this biogeographic imbalance to palaeoclimate dynamics. When climate is the only constraint to species' dispersal, our simulations yield more South-to-North than North-to-South dispersal events, mostly within the tropical biome. Also, our models fail to explain the migrant diversity desequilibrium between the two continents, as well as temperate specialists' over-representation in the fossil record of successfully-exchanged taxa. Hence, we argue that spatio-temporal palaeoclimate dynamics cannot explain the asymmetry of the GABI, suggesting additional factors fashioned this macroecological enigma.

Introduction

The Great American Biotic Interchange (GABI) refers to the large-scale biotic exchange that occurred between the Americas and intensified after the closure of the isthmus of Panama (Marshall *et al.* 1982; Stehli & Webb 1985; Webb 1976). Though the exact timing of the appearance of such a landbridge is not entirely resolved (Bourke *et al.* 2025; Jaramillo 2018; Montes *et al.* 2015; O'Dea *et al.* 2016), palaeontological evidence and molecular phylogenies both agree with a late Miocene onset of (non-volant) biotic exchange between the Americas (Carrillo *et al.* 2015; Tedford *et al.* 2004; Verzi *et al.* 2008), with effective exchange pulses occurring during the Plio-Pleistocene (Bacon *et al.* 2015, 2016; Marshall *et al.* 1982; Woodburne 2010). Among land mammals, a puzzling feature of the GABI is that it impacted both continental faunas very differently. In fact, evidence from the fossil record indicates that the early phases of the GABI were marked by reciprocal land mammal exchanges between the Americas (Marshall *et al.* 1982; Webb 1976; Woodburne 2010). However, during the Pleistocene, the ability of migrant lineages to maintain and diversify in the newly-colonised continent became greater in South America (Woodburne 2010), and many native South American lineages concomitantly declined, some of which even going totally extinct (Boscaini *et al.* 2025; Carrillo *et al.* 2020; Cione *et al.* 2015; Croft *et al.* 2020; Webb 1991). Hence, this intercontinental faunal exchange was asymmetric.

Although this macroecological imbalance is unquestionable, no clear consensus regarding the drivers underlying this asymmetry has been established yet. On the one hand, biotic interactions involving migrant and native species have often been proposed to explain such an asymmetry. These would include competition (Domingo *et al.* 2020; Simpson 1980; Webb 1976) or predation of predator-naive preys (Faurby & Svenning 2016; Fernández-Monescillo *et al.* 2025; Marshall & Cifelli 1990; Webb 1976). However, their contribution to the asymmetry of the GABI has been increasingly called into question (*e.g.*, Freitas-Oliveira *et al.* 2024a; Marsà *et al.* 2025; Prevosti *et al.* 2013; Tarquini *et al.* 2022; Vrba 1992). On the other hand, several studies also proposed the role of abiotic factors – *i.e.*, related to species' physical environment, such as climate (*e.g.*, Bacon *et al.* 2016; Freitas-Oliveira *et al.* 2024b; 2025; Marshall & Cifelli 1990; Vrba 1992; Webb 1991). The prevailing views consist of relating pulses of transcontinental mammal exchanges with Pleistocene climatic oscillations, promoting the transition from tropical rainforest to savannah-like environments in Central America as climate cooled down (Figure 1). Motivated by the migrant fossil record across both sides of the isthmus of Panama, Webb (1991) proposed that this climate-driven landscape opening would have facilitated the dispersal of open-adapted taxa. As those were assumed to be more numerous in North than in South America – the former continent offering more high-latitude temperate-to-cold environments than the latter –, a larger number of North American taxa would be predicted to disperse to the South than the other way around (Figure 1 A). The resulting exchanges would have been periodically interrupted as climate warmed up, promoting the reverse transition to rainforest in the isthmian region, with rainforests acting as a barrier for species dispersal (tropical 'holding pen' hypothesis) (Woodburne 2010). In addition, Vrba (1992) proposed the contraction of tropical habitats towards the equator promoted by the aforementioned periodical climatic cooling to have mechanically attracted taxa suited to tropical environments southwards (Figure 1 B). However, studies supporting these views have only been descriptive, tackling the issue from a pattern-oriented perspective driven by fossil data but, to our knowledge, never from a process-oriented perspective.

Recent improvements in process-explicit modelling have marked a turning point in the study of biodiversity (Pilowsky *et al.* 2022). By combining a landscape including palaeoenvironmental variables and a set of eco-evolutionary rules according to which virtual species are simulated, mechanistic eco-evolutionary models offer the unique opportunity to causally explore how biodiversity patterns arise and are maintained through evolutionary times (Hagen 2023). Recently, this class of models has provided unprecedented insights into the underlying forces behind widely-recognised biodiversity patterns, such as the latitudinal biodiversity gradient (Hagen *et al.* 2021a; Lorcery *et al.* 2025), the diversity desequilibrium between tropical rainforests across the globe (Hagen *et al.* 2021b), the build-up of the unique biodiversity along the Wallace line (Skeels *et al.* 2023b) and many others (*e.g.*, Boschman *et al.* 2023; Liu *et al.* 2024; Skeels *et al.* 2023a).

Here, we test Webb and Vrba's hypotheses on the role of palaeoclimate dynamics in the asymmetry of the GABI (Figure 1) using a spatially-explicit mechanistic biodiversity simulator coupled with palaeoclimate reconstructions across the last 5 million years. Our simulations tracked the diversification of a clade originating either in North or South America. The ability of a virtual species to maintain at a given place was only mediated by climatic factors. We hypothesise that (i) simulations starting from a North American ancestor would be more successful at reaching the other continent than the other way around, (ii) successful simulations starting from a North American ancestor would result in a more diverse and spatially spread radiation in the newly-colonised continent, and (iii) successful interchange would be more frequently achieved among biome generalists, extratropical specialists rather than tropical specialists.

Material and Methods

Beforehand, it has to be noticed that, across the entire study, "North America" actually refers to the current North and Central American continents. The delimitation between North and South America was arbitrarily set as a straight line connecting two points located along the eastern and western coasts of present Panama (82.882° W, 7.138° N and 81.369° W and 9.008° N) (see Figure S1).

Palaeoclimate data

We used the spatio-temporal terrestrial palaeoclimate series from the PALEO-PGEM atmosphere-ocean general circulation model (Holden *et al.* 2019), covering the last 5 Million years (My) with a 1 × 1 ° spatial resolution (Barreto *et al.* 2023). For the scope of this work, we used two bioclimatic variables: mean annual temperature (MAT, BIO1) and mean annual precipitation (MAP, BIO12). To avoid reaching an overkillingly-high number of simulation steps, we reduced the temporal resolution of the climate simulations from a 1000-years to a 10,000-years time step.

Additionally, in order to discriminate simulations based on the biome where the initial species originates, we reclassified monthly palaeoclimate layers for our starting time ($t = 5$ My ago) according to Köppen-Geiger broad climate types (Beck *et al.* 2018; Galván *et al.* 2023). This classification divides the world into five broad climatic regions: Polar, Cold, Temperate, Arid and Tropical, following the criteria proposed by Köppen (Köppen 1900).

Simulation model and modelling schemes

We performed spatially-explicit simulations of biodiversity using gen3sis v1.5.11 (Hagen *et al.* 2021a). Simulations, moving forward-in-time with a 10ky time step, tracked the radiation of a clade arising from a single ancestral population, either originating in North or in South America, throughout the last 5 My. In gen3sis, virtual species are modelled with individual populations evolving independently, following a set of four rules: dispersal, speciation, environmental filtering and trait evolution. All simulation scenarios that we implemented only differed on the last rule (trait evolution), the first three being common to all models (or at least partly).

We elaborated a gridded landscape relying on the two time-series of the aforementioned palaeoclimatic layers. Within this landscape, populations making up virtual species inhabited a set of connected cells. At each time step, inter-cell dispersal could occur with an intensity sampled in a Weibull distribution. If, by chance, two populations happened to be disconnected, they would accumulate divergence as long as they get disconnected, at a rate ruled by a divergence factor, here set to 0.05 (Table S1). Once passed a divergence threshold, speciation would occur – note that, consequently, the only mode of speciation allowed by gen3sis is allopatry.

As our goal was to evaluate the extent to which palaeoclimate change may have shaped the unequal success of North American taxa establishing southwards compared to South American taxa establishing northwards, local climate was set as the only constraint for a population to maintain in a given cell. Each species was therefore assigned a two-dimensional climatic niche, one axis corresponding to temperature suitability and the other to precipitation suitability. The abundance of a species along an environmental gradient was ruled by two parameters – that we also call traits – : a population- and species-specific optimum (resp. $P_{k,l}$ and $T_{k,l}$ for the precipitation and temperature optima of the population k of the species l) and a breadth (or tolerance) (resp. ω_P and ω_T; within a simulation, breadths were shared across populations and species). A population could maintain in a given cell only if the difference between its optima and local climate (both temperature and precipitation) was below its breadths. Therefore, given a site i with local precipitation and temperature at time t being $P_i(t)$ and $T_i(t)$, this translates in

$$\begin{cases} |P_i(t) - P_{k,l}| < \omega_P \\ |T_i(t) - T_{k,l}| < \omega_T \end{cases}$$

Each simulation started with a single ancestral species. The range of this ancestral species was randomly selected within the desired ancestral area (*i.e.*, North or South America). To avoid oversampling high latitudes, we merged the ancestral landscape with the centroids of an equal-area grid (~100km spacing) generated using the R package *h3jsr* (O'Brien 2023). We then sampled a centroid at random and retrieved the corresponding cell in our landscape (Figure S5 *left* and *right*). To reduce the probability of initial crash, the climate optima parameters of the ancestor were set to those of the local environment. However, due to the geometry of the two continents, this sampling scheme resulted in ancestral species starting from South America originating on average closer to the isthmus of Panama than those from North America (Figure S2 *top*). To control for this spatial bias, we assessed the distance to the isthmus of each cell of the North American portion of our 1 × 1° landscape. We then carried out other batches of simulations starting from North America, constraining the initial distance to the isthmus of the ancestral species of the simulation i to be as close as possible to the same distance involving the ancestral species of the simulation i starting from South America (Figure S2 *bottom*, Figure S5 *middle*).

500 replicates were made under each modelling scheme (detailed in the next paragraph), both with North and South America as ancestral area. Across replicates, we varied 6 key parameters: (1) the divergence threshold above which speciation occurred (S, range = $[1,3)$), (2) the rate of environmental niche evolution – only in models where we allowed climatic niche to evolve (σ_e, $[0.005,0.15)$), (3) the shape (ϕ, $[0.5,3)$) and (4) scale (ψ, $[0.1,1)$) of the Weibull distribution acting as a dispersal kernel, (5) the strength of environmental filtering (*i.e.*, niche breadth) for temperature (ω_T, $[0.01,0.1)$) and (6) precipitation (ω_P, $[0.05,0.3)$). As done by many authors, we ensured an even sampling of the resulting multidimensional parameter space via a quasirandom sampling technique relying on Sobol' sequences

[*e.g.*, Hagen *et al.* (2021b); Skeels *et al.* (2023b)]. Parameter ranges were determined on the basis of a literature review, ensuring their biological relevance. These ranges were further adjusted to limit the prevalence of crashing simulations. In addition to these varying parameters, we specified the value of several fixed parameters (Table S1).

We implemented various simulation scenarios (also called modelling schemes) where we modulated the ability of species to cope with climate change, and thus, the generalist vs. specialist nature of the simulated species. In the full-adaptationist model (*M1*, generalist), climatic optima ($P_{k,l}$ and $T_{k,l}$) were able to evolve in the direction of local climate change. To isolate the effect of a single environmental covariate on the resulting simulated biodiversity pattern, alternative versions of this model consisted of either removing the temperature (*M2*) or the precipitation (*M3*) constraint on species ecology. Besides, we also implemented a zero-adaptationist model (*M0*, specialist), where niche optima could not evolve and were therefore held constant (to their initial value) across all simulations.

Last, it is important to notice that, since the palaeoclimate reconstructions that we used ignored tectonic plate motion – a reasonable assumption for a 5-My time frame – continental position in our landscapes was held constant across simulations. Therefore, in that framework, changes in connectivity between North and South America could only arise from the environment.

Quantification and statistical analyses

At every time step, for each run of each simulation, we used a function released by Skeels *et al.* (2023b) to save presence-absence matrices. Given a time t, such matrices stored on their columns the presence of each simulated species that have existed from the onset of the simulation to t within each cell of the landscape (rows of the matrix) as a binary variable – 1: the species inhabits that cell, 0: it does not.

Given a batch of simulations, after discarding those that crashed – *i.e.*, that led to no living representative – we quantified four metrics reflecting the occurrence, magnitude and efficiency of an eventual transcontinental exchange based on the presence-absence matrix of the last time step of each successful run (hereafter called t_{end}). First, we assessed the exchange success, a binary variable set to one if at least one pixel of the area to colonise (hereafter called A_{col}, *e.g.*, South America if the simulation started in North America) recorded a presence. Then, we recorded the biome in which the initial ancestor occurs (*i.e.*, Tropical, Arid, Temperate, Cold, Polar) as well as the distance of this initial ancestor to a point arbitrarily located on the zone of the isthmus of Panama, (-84°lon, 10°lat). Last, in the case of a successful exchange, we quantified the number of virtual taxa recorded in A_{col} at t_{end} – referred to as "colonised diversity" – and the proportion of A_{col} actually colonised at t_{end} as the ratio between the number of cells occupied by the colonisers and the total number of available cells (*i.e.*, surface of A_{col}).

Results

South-to-North dispersal events are made easier by climate

Our results suggest that the proportion of simulations successfully reaching the other continent is always higher – by approximately a factor of two – when starting from South America than from North America (Figure 2). This imbalance holds whether we allow species climatic niche to evolve following the direction of local climatic change ($44 \pm 6\%$ vs. $87 \pm 3\%$, model $M1$) or not ($29 \pm 6\%$ vs. $59 \pm 5\%$, $M0$), although we can notice that overall transcontinental exchange frequencies are lower in the latter case. When allowing climatic niches to evolve, amputation of either their temperature ($47 \pm 7\%$ vs. $88 \pm 3\%$, $M2$) or precipitation ($38 \pm 4\%$ vs. $89 \pm 3\%$, $M3$) compounds did not alter the pattern. This imbalance was also maintained when standardising initial distances to the isthmus of Panama for simulated species originating in North America to those of simulated species from South America, except for simulations generated under the zero-adaptationist model ($50 \pm 6\%$ vs. $59 \pm 5\%$) (Figure S4).

Out of the 500 runs carried out for each initial landmass-modelling scenario combination, the proportion of simulations leading to no living representative (=crash) was on average three times higher for simulations starting from North America, except for model $M3$, recording almost no crash (Table S2). This difference, although attenuated, persisted after the spatial standardisation procedure.

Successful establishment on the newly-colonised continent

Irrespective of the initial continent, our findings highlight that successful simulations generated under models allowing species climatic niches to evolve ($M1-M3$) colonised a larger area on the newly-colonised continent than those generated under the $M0$ model – where climatic niche evolution was not allowed (Figure 3 A). Furthermore, species of North American origin that successfully made it to South America generally colonised a greater area than those of South American origin reaching North America (Figure 3 A). This pattern is particularly striking for simulations generated under the $M0$ model ($36.5 \pm 3.5\%$ vs. $11.7 \pm 0.4\%$), and holds for all but the $M3$ model – relaxing the precipitation constraint on species' niche –, the latter suggesting an opposite trend ($69.9 \pm 3.3\%$ vs. $79.4 \pm 3.9\%$). However, due to the difference in size (*i.e.*, number of pixels) between the two continents (Figure S1), the actual colonised area was greater when simulations generated under the $M1-3$ models started from South than North America, but remained larger when simulations were generated under $M0$ started from North America (Figure S11).

Besides, contrary to the proportion of colonised area for a given ancestral continent, colonised diversity does not vary significantly across our four models. However, a significant decrease is evident in simulations starting from North America generated under the $M3$ model (Figure 3 B). Moreover, ancestral continent does not seem to affect mean colonised diversity values in the case of the $M1$ (212 ± 61 vs. 170 ± 25 species respectively retrieved in the newly-colonised continent after starting from North or South America) and $M2$ (161 ± 48 vs. 171 ± 27 species) models. It does however have an effect for simulations generated under $M0$, with higher average values obtained for simulations starting from North America (247 ± 85 vs. 79 ± 20 species).

In addition, successful simulations starting from North America generally started from an ancestor located closer to the isthmus of Panama than those starting from South America (Figure 3 C), again except for simulation generated under $M3$. Also, we can highlight a general trend that ancestral species originated closer to the isthmus in the case of successful simulations compared to unsuccessful ones (Figure S7-8). Interestingly enough, the results obtained for this metric, just like those obtained for the other two, hold for simulations generated after standardising the initial distances to the isthmus of North American virtual species with those starting from South America (Figure S4).

Biome selectivity of the simulated faunal exchanges

Due to the unequal area occupied by each biome between the two continents, the number of simulations starting within each biome was largely uneven across biome classes and continents (Figure 4 B). The majority of North American ancestral species originated in the cold biome (57%), whereas the tropical biome recorded the largest proportion of ancestral species origination in South America (58%). Standardising initial distances to the isthmus of Panama for simulations originating in North America increased the proportion of simulations originating under low latitudes, to the detriment of those originating in high latitude, less represented in South America (Figure 4 A and Figure S5). Consequently, this increased the representativity of the tropical (from 5% to 10%) and arid (from 16% to 26%) biomes in the starting ranges of simulations originating in North America, and decreased the proportion of simulations originating in the cold (from 57% to 40%) and polar (from 4% to 2%) biomes (Figure S10).

Despite this initial imbalance, when simulations were generated under the $M0$ model, our results suggest similar exchange intensity patterns between simulations starting from North and South America (Figure 4 C, first row). Tropical specialists (*i.e.*, originating in the tropical biome and evolving under $M0$) were largely more frequently exchanged than other biome specialists, with on average $90[77,100)\%$ successful exchanges southwards (*i.e.*, from simulations originating in North America) and $66 \pm 5\%$ northwards (*i.e.*, from simulations originating in South America). Such proportions were significantly reduced for arid ($26 \pm 6\%$ southwards, $18 \pm 9\%$ northwards) and temperate specialists ($26 \pm 10\%$ southwards, $29 \pm 8\%$ northwards), and almost no cold or polar specialist, no matter where they originated, successfully made it to the other continent. The exact same pattern is obtained for southwards dispersal events occurring after standardising initial distances to the isthmus of Panama (Figure S9, $M0$).

Regarding biome generalists (models $M1-3$), simulations starting from North America maintained the same distribution of exchange intensities across biomes as biome specialists. Generalists that originated in the tropical biome record the highest proportion of exchange success ($89[75,100)\%$ for $M1$), followed by those originating in the temperate ($40 \pm 10\%$, $M1$) and arid ($34 \pm 11\%$, $M1$) biomes. Also, generalists originating in the cold or polar biomes almost never made it to South America (Figure 4 C $M1-2$). Note that removing the precipitation constraint on species climatic niche (*i.e.*, model $M3$) attenuated this imbalance (Figure 4 C $M3$). All this still holds when standardising initial distances to the isthmus of Panama of simulations starting from North America to those starting from South America (Figure S9, $M1-3$). Conversely, the disparity in proportion of successful exchanges across biomes was largely attenuated for biome generalists originating in South America, for which a successful exchange appears almost equally-likely, regardless of the biome where they originate (Figure 4 C $M1-3$).

Discussion

An inverse asymmetry of exchange driven by climate

Throughout this work, we investigated the role of palaeoclimate in the asymmetry of the GABI through the use of eco-evolutionary simulations. Probably one of our most striking findings is that, under climatic constraints alone, simulated South American lineages dispersed to the North with a significantly greater intensity than the other way around (Figure 2). This therefore clearly contrasts with the long-recognised imbalanced success of North American mammal taxa compared to South Americans in the aftermath of the GABI (Simpson 1980). Seminal works primarily interpreted the empirically-derived asymmetry of the GABI under the lens of a continent-wide extension of MacArthur & Wilson's equilibrium theory (MacArthur & Wilson 1967). The latter theory establishes that, on evolutionary timescales, diversity within a given area should tend towards an equilibrium value proportional to the size of this area. Therefore, its continent-wide extension predicts the equilibrium diversity of North America to be larger than that of South America, given the differences in size between the two continents (24 vs. 18×10^6 km²). Following this framework, Marshall *et al.* (1982) argued that the continent offering the largest pool of taxa – which, according to this prediction, would be North America – would in turn have resulted in the larger number of species exchanged in the end. It might be noticed that species pool imbalance between two source areas has been identified as a key driver in the faunal exchange asymmetry that contributed to the build-up of the unique biodiversity along the Wallace line (Skeels *et al.* 2023b). Nonetheless, following such a reasoning, the probability for a given taxon to cross the Panamanian landbridge is assumed to be independent from the side of the isthmus where it came from, which, interestingly, is not supported by our findings (Figure 2). In addition, this theory does not align with the actual imbalance in extant mammal species-level diversity between the Americas, being way larger in South than in North America (Raven *et al.* 2020).

Although the inverse asymmetry in climate-driven exchange across both sides of the Isthmus suggested by our findings goes in contradiction with the mammal fossil record (*e.g*., Marshall *et al.* 1982), it actually echoes inter-American exchange patterns suggested by other groups of vertebrates. For instance, the extant South American amphibian fauna is almost entirely endemic (Lomolino *et al.* 2006). By contrast, the North American frog fauna is composed of four families of southern origin, and only three of northern origin (Lomolino *et al.* 2006). Just like herpetofauna, tropical North American ichtyofauna has long been recognised as mainly composed of groups that originated in South America and further dispersed to the North, although most likely multiple dispersal events occurred, some of which pre-dating the closure of the isthmus (*e.g*., Chakrabarty & Albert 2011; Lomolino *et al.* 2006; Smith *et al.* 2012). It should however be noted that, due to the requirement for connected freshwater networks, the magnitude of the faunal admixture among freshwater fishes has been greatly limited compared to land groups. Next, despite the higher dispersal abilities of birds, phylogenetic evidence has shown a significant influence of the closure of the Panamanian Seaway in the rates of exchange between both continent's avifaunas (Smith & Klicka 2010; Weir *et al.* 2009). In particular, these studies suggested that transcontinental exchanges occurred symmetrically among clades inhabiting tropical regions, but interestingly, just like land mammals, diversification of extra-tropical groups to tropical regions and further dispersal to the other landmass was strongly asymmetrical in favour of North American groups (Smith *et al.* 2012; Smith & Klicka 2010). That being said, notable exceptions of South American bird groups that successfully diversified in North America (including within temperate regions) exist, such as hummingbirds (*e.g*., Wallace 1878). However, the lack of fossil record for all the aforementioned groups may hamper the understanding of the processes behind their biogeographic history. Last, the Central American flora has been characterised as a mixture of lineages of North and South American origins, but phylogenetic evidence has suggested that the arrival of taxa from South America largely predated the established timeline of the GABI (*e.g*., Cody *et al.* 2010). Nevertheless, the orchid flora of the Chocó region, bridging the north-western coast of South America and southern Panama, was found to be primarily made up with Andean − thus, South American − lineages that radiated locally not earlier than 5 million years ago (Pérez-Escobar *et al.* 2019).

Revisiting climatic paradigms of the Great American Biotic Interchange

One of the predictions of Vrba's resource-use hypothesis, formulated within the framework of her Habitat Theory and applied to the GABI, is that tropical specialists would have been naturally attracted towards the equator following the contraction of the tropical belt in the aftermath of the global cooling occurring at the Plio-Pleistocene boundary (Figure 1 B; Figure S1, from *ca*. 3 Ma). This would have actively promoted the establishment of North American tropical specialists in South America (Vrba 1992). Here, our virtual experiments partially corroborate these historical predictions. Indeed, simulated tropical generalists of North American origin – coming from simulations starting in tropical North America and of which climatic niche could not evolve, model $M0$ – were found to have a high prevalence of crossing the isthmus (Figure 4 C, model $M0$). However, the prevalence of tropical specialists from South America to disperse to the North was found to be almost equally-high, largely exceeding that of other biome specialists (Figure 4 C, model $M0$). This goes against Vrba's seminal hypothesis, instead predicting that fewer mammals – only biome generalists or open-habitat specialists – from South America would be expected to roam the other way around in case of tropical contraction towards the equator. This finding could reflect an attenuated magnitude of the climatic fragmentation of the Panamanian region when compared to historical biome reconstructions (*e.g.*, Webb 1991), with the persistence of a tropical corridor in turn promoting the exchange of tropical specialists between both landmasses.

On average, one simulated radiation of open biome – *i.e.*, arid or temperate – specialists out of four resulted in a successful exchange, irrespective of its ancestral continent (Figure 4 C, model $M0$). This proportion is significantly lower than that obtained for tropical specialists. Yet, it has long been acknowledged that the migrant fossil record across both sides of the isthmus was essentially composed of taxa showing affinity with open biomes (*e.g.*, Vrba 1992). Among the most emblematic, we find for instance equids, proboscids or camelids on the South American side, and ground sloths, toxodontids or glyptodontids on the North American side. The co-occurrence of distinct taxa in temporally well-delimited fossil assemblages led palaeontologists to postulate that the GABI was in fact a succession of exchange pulses punctuated by phases of interruption, further related to the climate-driven oscillations between savannah and rainforest in Central America during the Pleistocene (Webb 1991). Under such an assumption, the majority of exchanged taxa would therefore have ecologies suited to open habitats, and would have preferentially dispersed during Pleistocene glaciations (Bacon *et al.* 2016; Woodburne 2010) (Figure 1 A). As mentioned by Weir *et al.* (2009), the American mammal fossil record is highly biased, with few occurrences coming from low-latitude tropical regions, which is particularly striking in South America (*e.g.*, Ugarte *et al.* 2025). Furthermore, evidence from extant mammal taxa suggest a strong tropical niche conservatism, that is, the trend according to which newly-originated tropical taxa would remain inhabiting under tropical climates (Smith *et al.* 2012; Wiens & Donoghue 2004). Thus, the fossil-derived number of mammal exchanges must have been greatly underestimated, with little fossil evidence of interchange constrained to the tropical biome, as stressed by Carrillo *et al.* (2020). Thus, one can wonder: could the empirically-derived migration imbalance in favour of North American taxa dispersing southwards be altered by the tropical niche conservatism of South-to-North migrants, thereby obscuring the actual number of intercontinental exchanges?

Our findings revealed that radiations following a colonisation of South America were more spatially spread than those arising from a northwards dispersal, except for simulations generated under the $M3$ model (Figure 3 A). The spatial spread of a migrant radiation was measured by the proportion of cells occupied within the newly-colonised continent relative to the size of the latter. When focusing on the absolute number of cells colonised – therefore without controlling for the area, in terms of number of cells, of the newly-colonised continent – such a finding only holds for successful simulations generated under $M0$ (Figure S11). This mirrors the fact that simulated tropical specialists from South America were more intensively exchanged than other simulated biome specialists (Figure 4 C, S6). The imposed tropical niche conservatism therefore constrained them to remain in tropical regions of North America, that are less represented in tropical North America than in South America (Figure 4 A). This adds some evidence supporting the fact that climate would have permitted South-to-North tropical exchanges to take place, but might have been obscured by a subsequent taphonomic bias in the fossil record, as suggested by studies relying on dated phylogenies (Bacon *et al.* 2015; Carrillo *et al.* 2020). Hence, our study calls for thinking about the distinction between the ability to disperse towards, and ability to maintain within the newly-colonised area. Those two distinct processes may have been ruled by different factors.

For additional remarks, limitations and perspectives, see *Supplementary text*.

In summary, our findings do not support the hypotheses exclusively relating the asymmetry of the GABI to palaeoclimate dynamics. Rather, our virtual experiments suggest that palaeoclimate would not have prevented reciprocal exchanges between the faunas of the two continents, especially within tropical areas. This stresses that the tropics did not act as a barrier as previously stated. Given the scarcity of fossil material in the tropical areas of the Americas, hypotheses relating the asymmetry of the GABI to palaeoclimate dynamics could have been affected by taphonomic bias, as they largely rely on the land mammal fossil record. This ultimately suggests that, although endemic land mammals from South America were virtually able to reach the other landmass, other unconsidered factors must have prevented their establishment on the other continent, in turn shaping the widely-recognised asymmetry of this textbook example of biotic interchange.

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We thank Adrián Castro Insua for his useful guidelines and recommendations at the onset of the project. We thank Céline Devaux for her useful feedback on the present work. L.B. was funded by a grant offered by the École Normale Supérieure de Lyon. This work received support from the European Research Council under the European Union's Horizon 2020 research and innovation program (grant agreement 947921) as part of the MAPAS project, and benefited from an “Investissements d’Avenir” program managed by the Agence Nationale de la Recherche (CEBA, ref. ANR-10-LABX-25-01).

398 # Figures

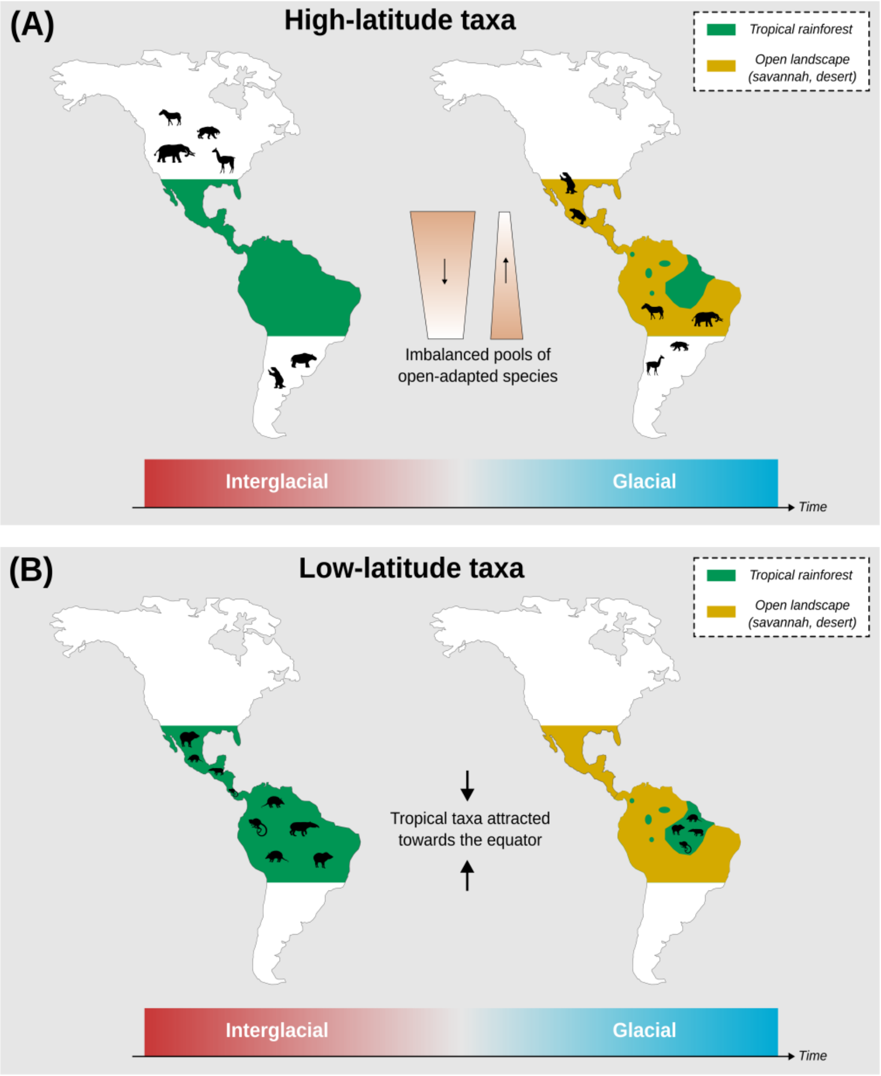

Figure 1: The two nonmutually-exclusive hypotheses relating the asymmetry of the GABI to the Pleistocene climatic oscillations, formulated in the framework of land mammals. Both involve biome transition from close tropical rainforest (green) during interglacial periods − warm and humid − to open savannah/desert (dark yellow) during glacial maxima − cold and dry − in the isthmian region. The first one (A), formulated by Webb (1991) views the asymmetry of the GABI as the result from the fact that North America had a higher pool of open-adapted taxa than South America. The opening of the isthmian region acted as an ecological filter favouring the dispersal of these open-adapted taxa to the detriment of the others. Hence, given the imbalance in open-adapted pool size between the two continents, more species would have been expected to disperse to the South than the other way around. The second one (B), formulated by Vrba (1992), proposes that the contraction of the tropical biome towards the equator triggered by the interglacial-glacial transition would have mechanically attracted tropical specialists southwards, irrespective of their continent of origin. Silhouettes are from phylopic, credits are available in Table S3.

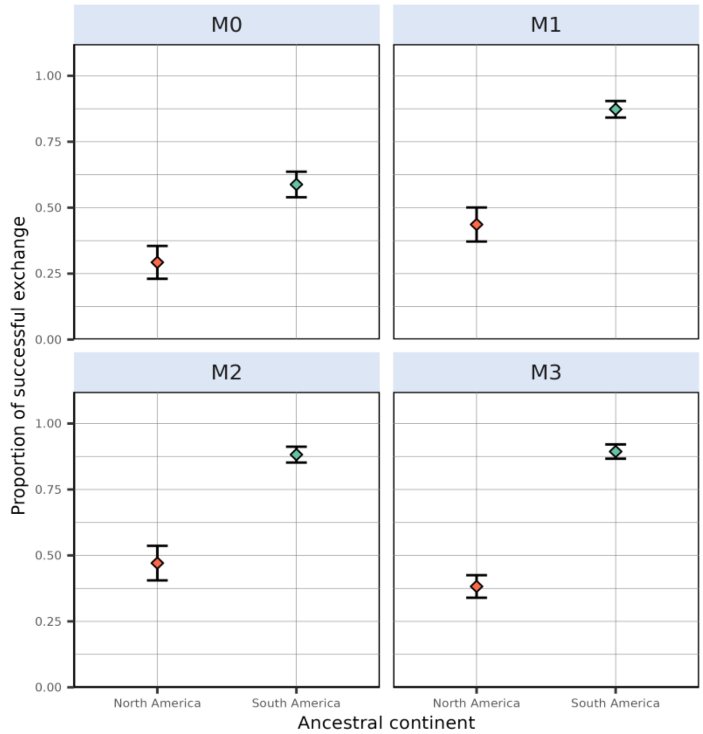

Figure 2: Proportion of simulations leading to a successful exchange to the other continent, computed as the ratio, given an initial continent and a modelling scenario, between the number of simulations leading to a transcontinental exchange and the total number of simulations resulting in at least one living species at the final state. From left to right are depicted the outcome of the zero-adaptationist model ($M0$, species climatic niche does not change through time – climate specialists), full adaptationist model ($M1$, species climatic niche evolves following local changes in temperature and precipitations – climate generalists) and alternative versions of $M1$ where species ecological niche is only determined by precipitation ($M2$) or temperature ($M3$). Each individual plot shows the outcome of simulations starting in North (red) and South (cyan) America. Dots represent mean proportion estimates and error bars show their associated Binomial-derived 95% confidence interval.

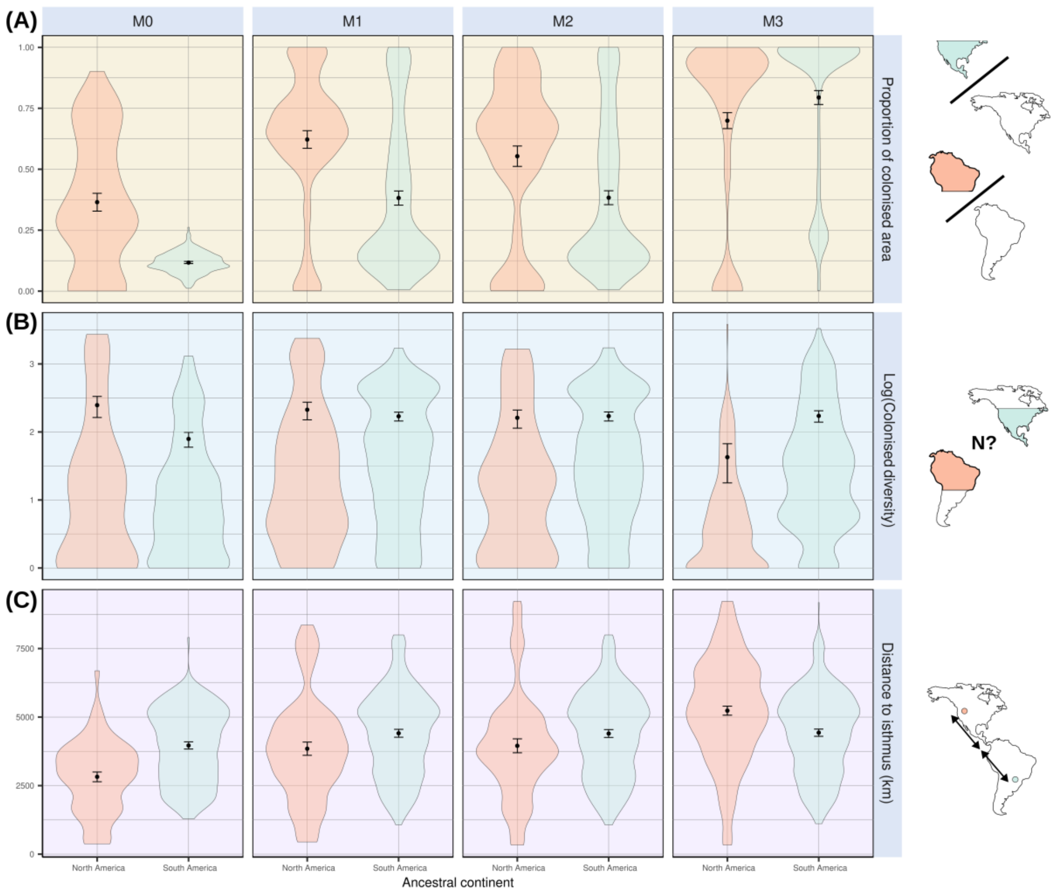

Figure 3: A few metrics summarising climate-mediated interchange derived from successful simulations, *i.e.*, that resulted in a transcontinental faunal exchange. From left to right are shown the results of simulations generated under our four models (see Figure 2 for details). **(A)** Proportion of newly-colonised continent's colonised area, quantified as the ratio between the number of pixels recording a presence in the landmass to colonise and the total number of pixels offered by this landmass. **(B)** Colonised diversity, expressed as the natural logarithm of the number of lineages within the colonised area. **(C)** Distances between the location where ancestral species originated and the isthmus of Panama, only for successful simulations – *i.e.*, successfully reaching the continent where it did not originate. Mean estimates and their associated Student-derived 95% confidence interval are displayed over their distribution.

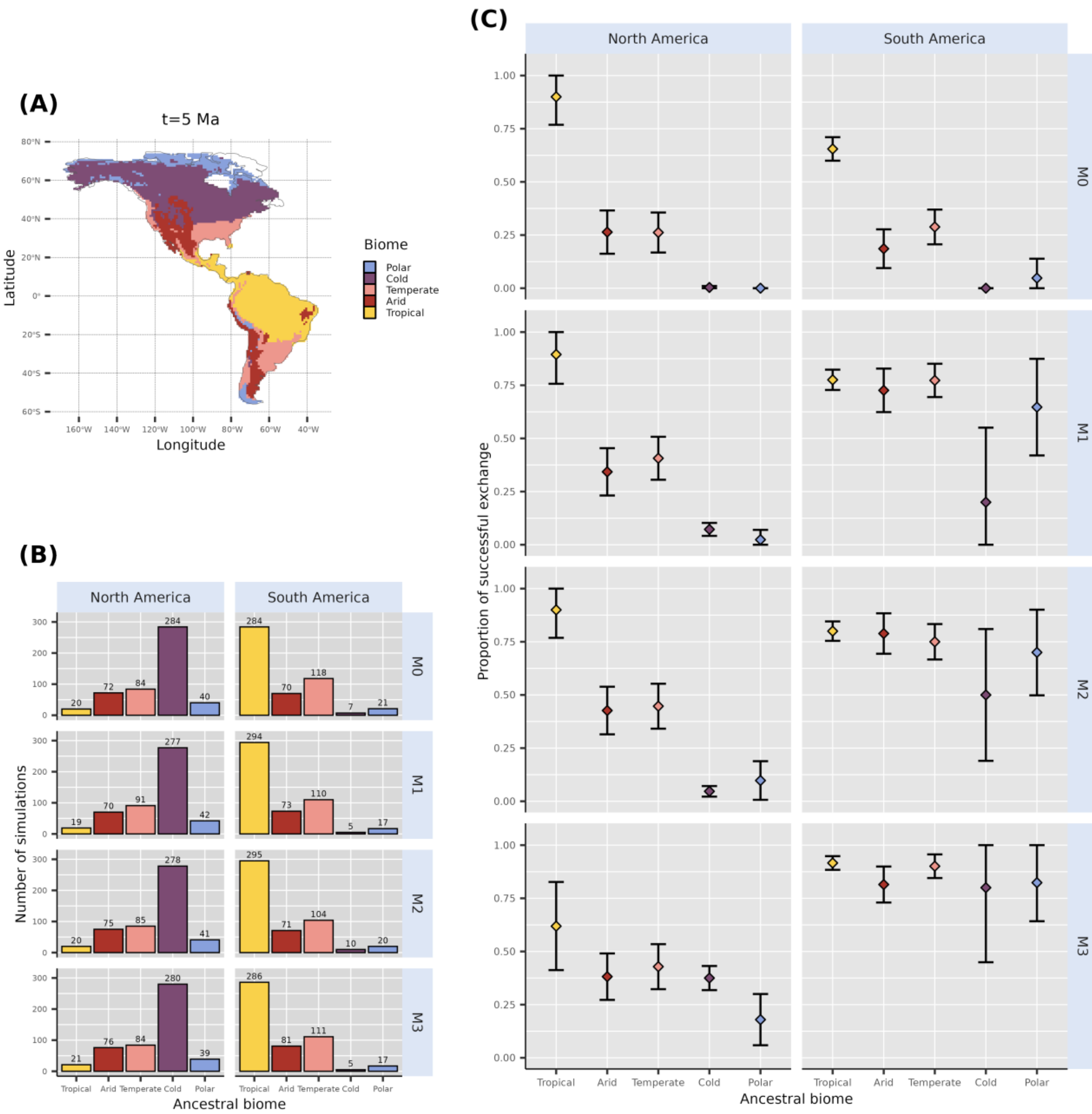

Figure 4: Initial biome where each ancestral species leading to successful simulations is recorded at the time of the onset of the simulations (5 Ma). **(A)** Global main climate zones based on Köppen-Geiger biome classification at $t = 5$ Ma. **(B)** Biome where the ancestral species of each of the 500 simulations originated, for each continent and each scenario. We removed simulations starting from a cell that could not be assigned a biome from our counts, hence the reason why some are not exactly summing to 500. **(C)** Proportion of successful simulations based on the biome where ancestral species originates, for each modelled scenario-ancestral continent combination. This proportion is expressed as the ratio between the number of successful simulations and the total number of simulations starting from a given biome. Coloured dots and error bars respectively show proportion estimates and their associated Binomial-derived 95% confidence intervals.

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Supplementary Materials for: Palaeoclimatic constraints behind the Great American Biotic Interchange: insights from eco-evolutionary simulations

Lucas Buffan[1], Fabien L. Condamine[1], Sara Varela[2]

[1] Institut des Sciences de l'Evolution de Montpellier, Université de Montpellier, CNRS, IRD, Place Eugène Bataillon, 34095 Montpellier cedex 5, France.
[2] Centro de Investigación Mariña, Departamento de Ecoloxía e Bioloxía Animal, Universidade de Vigo, Campus Lagoas-Marcosende, 36310 Vigo, Spain

Supplementary text

Geographic contingency and the Great American Biotic Interchange

Our findings leave room for possible random effects related to geography. Indeed, irrespective of the initial landmass, successful simulations originated on average closer to the isthmus of Panama than unsuccessful ones (Figures S6–7). This suggests that some simulations were unsuccessful not only due to climatic filters, but also because they simply started too far from the landbridge, preventing them from reaching it during the simulation. The relationship between a taxon's success in reaching the other continent and the distance between its geographic range and the isthmus of Panama has long been postulated in the literature (*e.g.*, Simpson 1980). Webb (1991) illustrated this view with the example of antilocaprids, which "were probably capable of living somewhere in the Andes, [but] were simply too far from the isthmian link to make the exodus". Following this line, a recent study characterised a negative correlation between the proportion of migrant taxa within fossil assemblages of either North or South America and the distance of the fossil assemblage to the Panamanian landbridge (Motta & Quental 2025). Therefore, geographic contingency must also have played a significant role in the asymmetry of the GABI.

Limitations and perspectives

Our work comprises a wealth of limitations. Below, we discuss some of them and propose possible avenues for future improvements.

Differences in the pace at which migrant and native lineages diversified have long been proposed to play a significant role in the asymmetry of the GABI. By reconstructing raw macroevolutionary dynamics from the fossil record, Marshall *et al.* (1982) highlighted that northern immigrant genera experienced notoriously higher origination and lower extinction rates than native lineages in South America. More recently, using a Bayesian framework accounting for preservation heterogeneities to infer fossil-based macroevolutionary trends, Carrillo *et al.* (2020) showed that, in their home continent, native South American lineages experienced higher extinction rates than migrant lineages. In our simulations, for a species to go extinct, it needs to disappear from each cell across the landscape. As environmental filtering – in our case, the only force driving diversification – acts at the scale of a pixel, species extinction is therefore very rare. As a result, our framework did not allow us to relate the simulated environment-dependent inter-American faunal exchange with any diversification process. Realistic modelling of macroevolutionary processes is therefore an avenue towards improvements among mechanistic eco-evolutionary simulators, particularly in the framework of the GABI.

Also, we evidenced a nuance between the proportion of colonised area and the absolute colonised area (*Main text*, Figure 3 and Figure S11) simply comes from the imbalanced number of pixels between the two landmasses, being around two times larger for North America than for South America (Figure S1). A possible avenue to avoid such confusions when carrying eco-evolutionary simulations at such a broad spatial scale (here from *ca.* 80° N to *ca.* 40° S lat) would be to use equal-area landscapes. In our case, this would require conversions in the palaeoclimate simulations that we use, as well as adapting dispersal probabilities based on actual distances rather than equal-degree pixels.

The main message of our study is that climate alone is not sufficient to explain the asymmetry of the GABI. Thus, other factors unaccounted here might have been at play to shape the uneven faunal exchange that occurred across the Americas during the GABI. Among them, one can think about biotic interactions. Convinced that the overwhelming success of North American mammals in South America could "*not be pure coincidence*", Simpson (1950) proposed the asymmetry of the GABI to stem from the different levels of exposure to competition between the North and South American faunas. Throughout the Cenozoic, exposure to competition had indeed been much larger for North than South American mammal taxa, due to the isolated charcter of South America, contrasting with the several connections of North America with other continents (*e.g.*, Eurasia). In addition, some studies proposed the asymmetry of the GABI to result from prey-predator interactions, with Northern lineages preying on Southern ones (*e.g.*, Faurby & Svenning 2016). However, several process-based analyses of the fossil record provided no evidence for biotic interactions involving native South American mammals and their northern counterparts (*e.g.*, Prevosti *et al.* 2013; Tarquini *et al.* 2022). Hence, incorporating biotic compounds in mechanistic frameworks such as the one we have implemented here would be a worthy avenue for future studies tackling the asymmetry behind the GABI.

Last, the landscape designed for the purpose of this study (Figure S1) only permitted transcontinental faunal exchange via the isthmus of Panama, the closure of which being assumed to be the main trigger of the GABI (Marshall *et al.* 1982). Yet, the fossil record of the Caribbean Islands has provided evidence for faunal supply both from the South and the North American landmasses, in particular since the late Miocene (MacPhee 2011; *e.g.*, Morgan 1993). Although over-water dispersal has long been regarded as the prevailing mechanism (Morgan 1993), recent geological advances have provided evidence for (1) the emergence of land bridges connecting South America to Caribbean Islands and (2) the formation of mega-islands – during Pleistocene glacial maximum episodes – several times since the Mio-Pliocene boundary Cornée *et al.* (2021). Hence, it would make sense to consider the Caribbean Islands as an alternative dispersal pathway between the Americas, as shown to be the case in earlier time frames, for instance, near the Eocene–Oligocene boundary (*i.e.*, GAARlandia hypothesis) (Cornée *et al.* 2021; Marivaux *et al.* 2020). However, due to the idiosyncratic eco-evolutionary processes shaping insular systems (*e.g.*, Warren *et al.* 2015) as well as the poor mention of the Caribbean Islands as a stepping-stone in the literature of the GABI, we believe that considering this alternative route in our framework would rather deserve to be the topic of another study.

Supplementary figures

All across the study, metrics computed out of simulations starting from North America are shown in coral red, and those starting from South America in cyan.

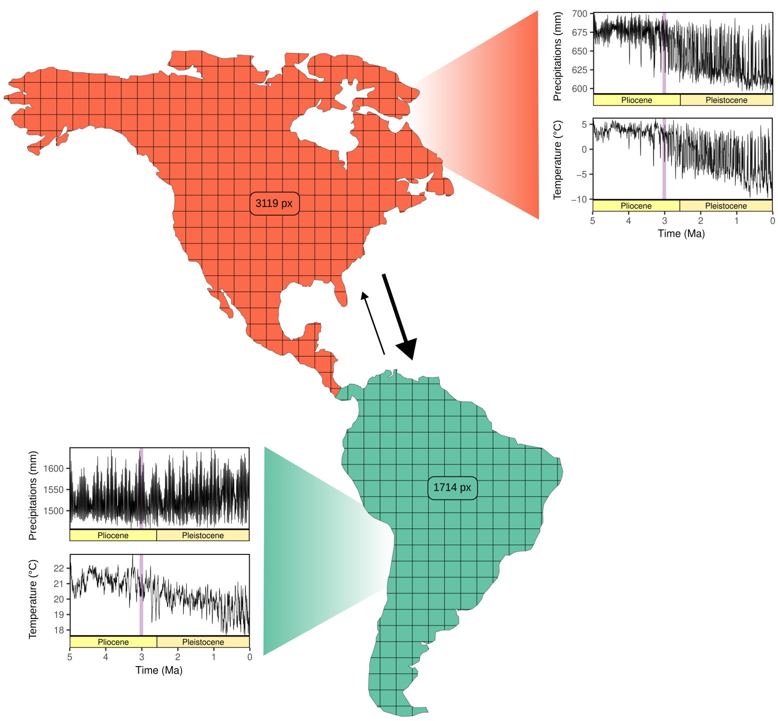

Figure S 1: Average palaeoclimate histories within our studied landmasses. North America is displayed in coral red and South America in cyan. Spatially-averaged climate variables (Annual Temperature and Precipitation) are taken from the PALEO-PGEM palaeoclimate emulator Holden *et al.* (2019). This emulator being spatialised at a 1 x 1° resolution, numbers inside the continents represent the quantity of overlaying pixels from the climate grid within each continent. On climate plots, the time frame of the onset of significant land mammal exchange across the americas (Bacon *et al.* 2016; Woodburne 2010) is indicated by the light purple vertical bands. The empirically-derived asymmetry of the transcontinental land mammal exchange is represented by the thickness of the two arrows, the thicker the arrow, the more intense the exchange.

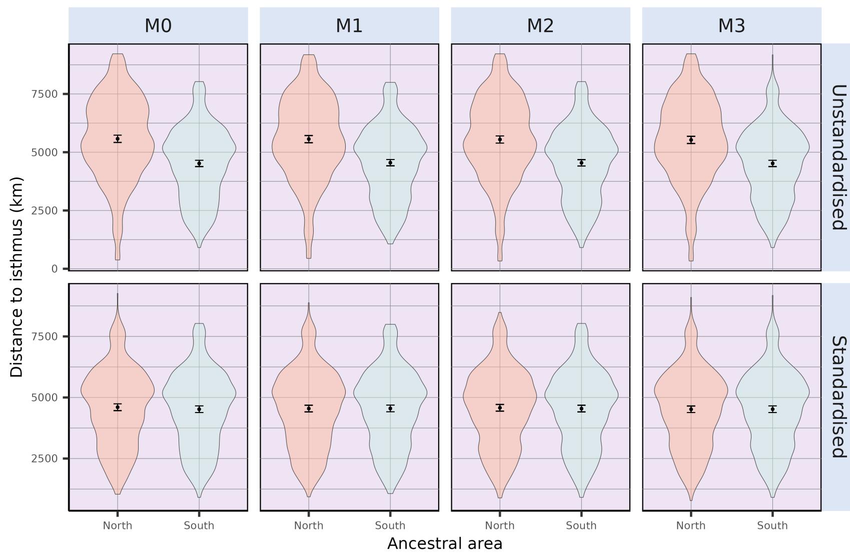

Figure S 2: Initial distances to the isthmus of all simulations ($n = 500$), either starting from North or South America. On the top row, locations of simlation's ancestral species resulted from a random landmass-scale equal-area sampling scheme, both starting from North and South America. On the bottom row, initial distances to the isthmus of simulations originating in North America were standardised to those of South America (see *STAR Methods*). Note that no change was made for simulations starting from South America.

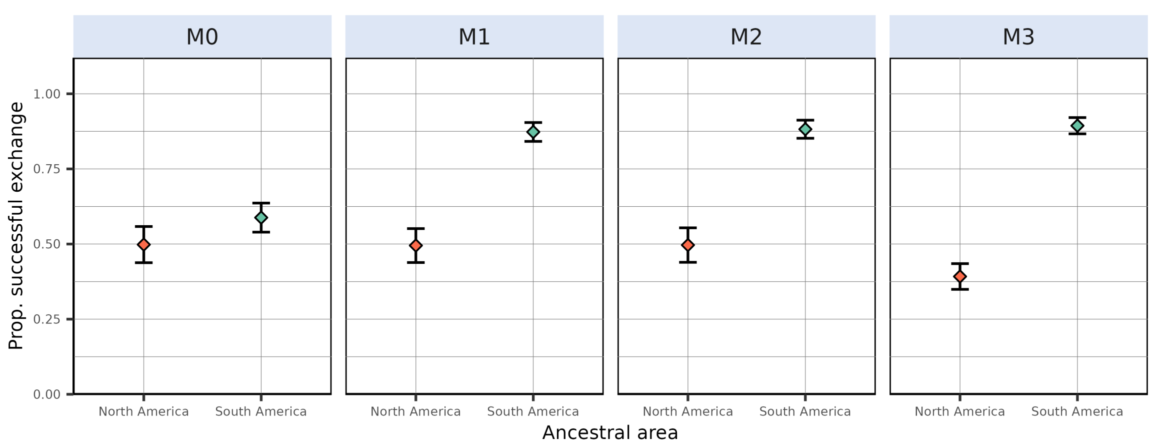

Figure S 3: Proportion of simulations leading to a successful exchange to the other continent after standardising initial distances to the isthmus of simulated species originating in North America to those of simulated species from South America (see *STAR Methods*). This proportion is computed as the ratio, given an initial continent and a modelling scenario, between the number of simulations leading to a transcontinental exchange and the total number of simulations resulting in at least one living species at the final state – *i.e.*, excluding crashes, see Table S2. From left to right are depicted the outcome of the zero-adaptationist model ($M0$, species climatic niche does not change through time – climate specialists), full adaptationist model ($M1$, species climatic niche evolves following local changes in temperature and precipitations – climate generalists) and alternative versions of $M1$ where species ecological niche is only determined by precipitation ($M2$) or temperature ($M3$). Dots represent mean success proportion estimates and error bars show their associated Binomial-derived 95% confidence interval.

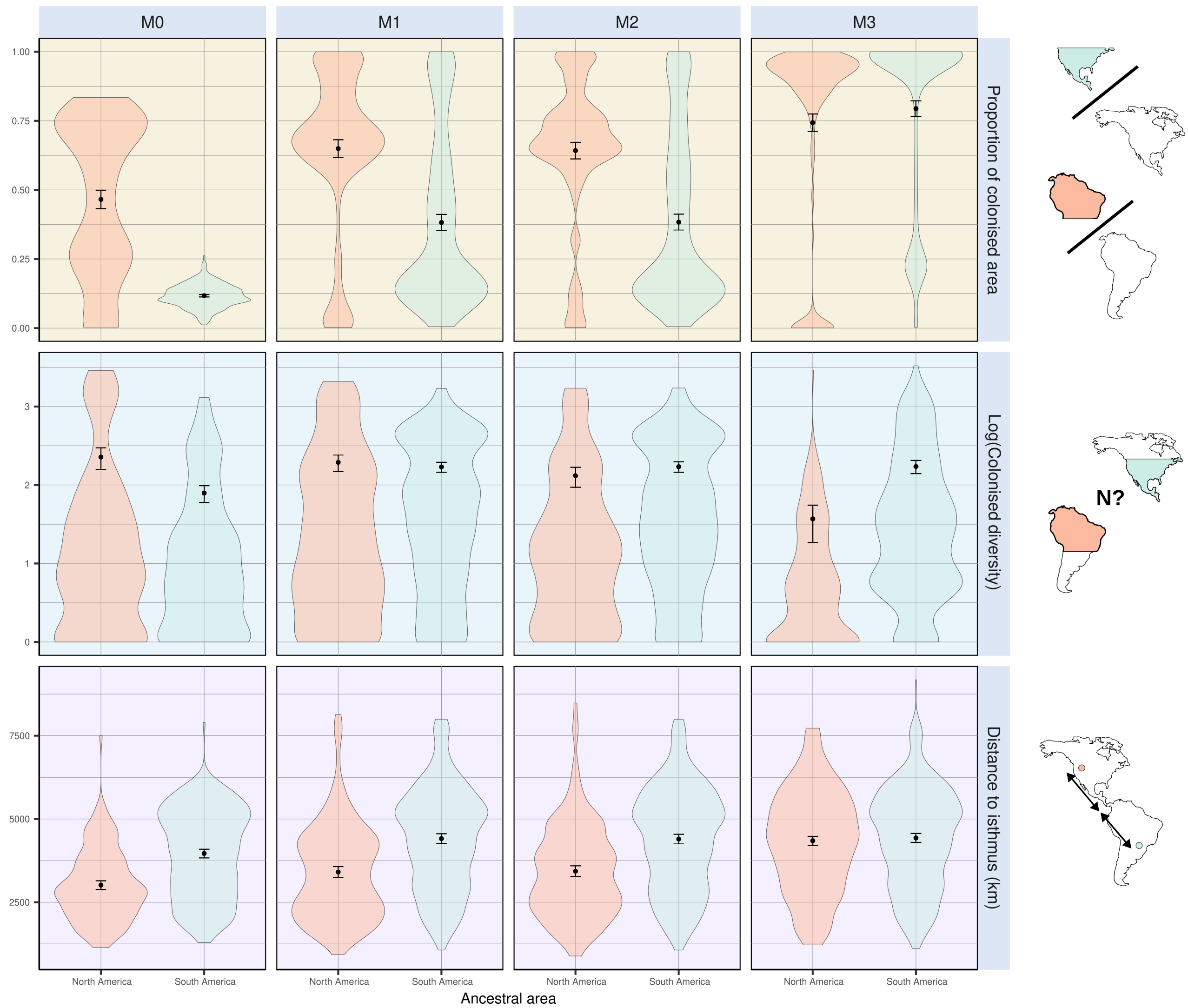

Figure S 4: A few metrics summarising climate-mediated interchange after standardising initial distances to the isthmus of simulations starting from North America to those starting from South America. Each individual plot shows the outcome of simulations starting in North (red) and South (cyan) America. (*top*) Proportion of foreign continent's colonised area. (*middle*) Diversity within the colonised area. (*bottom*) Distance to the isthmus. The first two metrics have been computed out of successful simulations, *i.e.*, that resulted in a transcontinental faunal exchange.

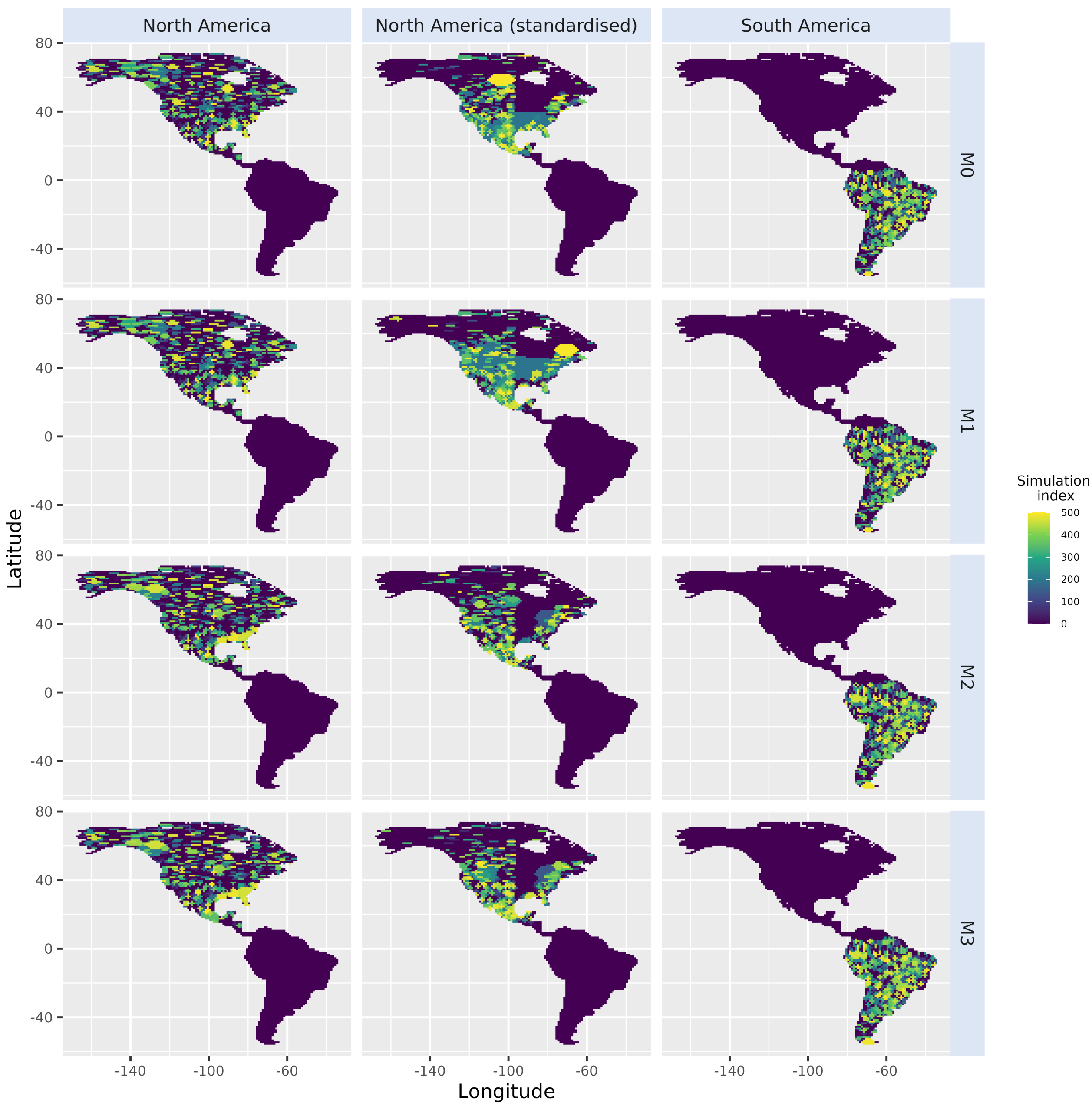

Figure S 5: Starting range of all simulations either starting from North America (*left*), North America after standardising initial distances to the isthmus with those from South America (*middle*) or South America (*right*).

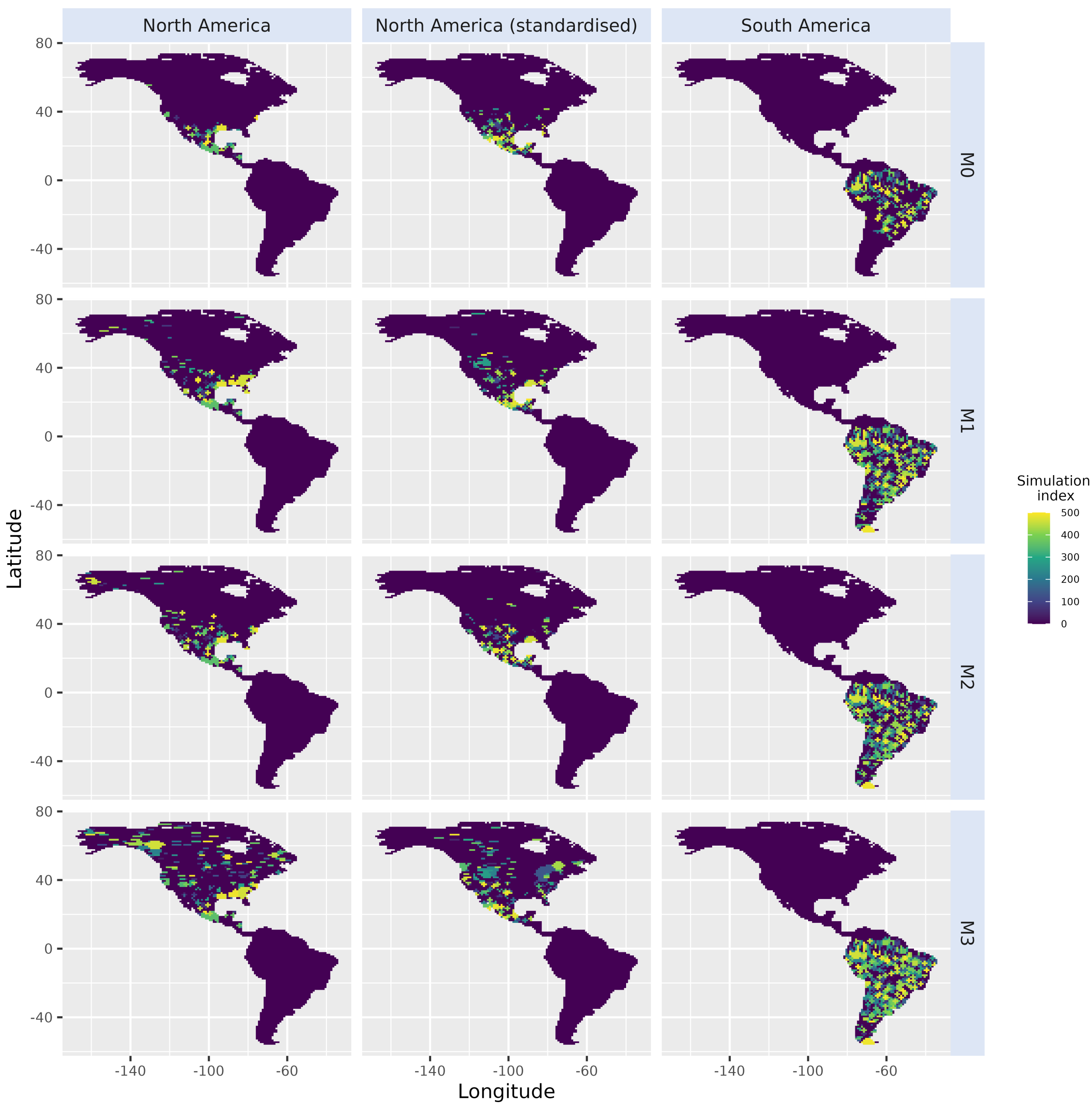

Figure S 6: Starting range of successful simulations generated under each model (from $M0$ to $M3$), either starting from North America (*left*), North America after standardising initial distances to the isthmus with those from South America (*middle*) or South America (*right*).

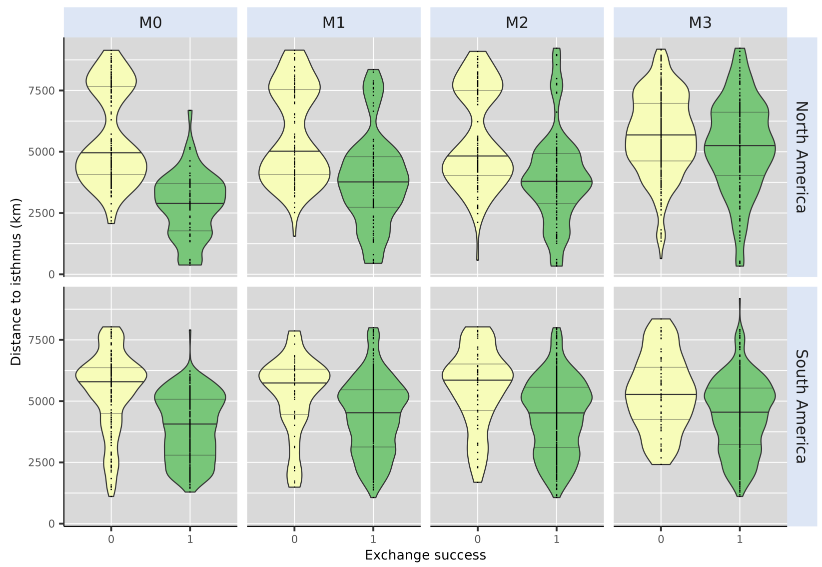

Figure S 7: Comparison of the initial distance to the isthmus between successful and unsuccessful simulations of each of our four models, either starting from North (top) or South (bottom) America. Successful simulations ($Exchange\ Sucess = 1$) are displayed in green, whereas unsuccessful simulations ($Exchange\ Sucess = 0$) are displayed in yellow. Dashed lines on top and bottom of the distribution represent the 25% and 75% quantiles whereas the thicker full line in the middle shows the median (50% quantile).

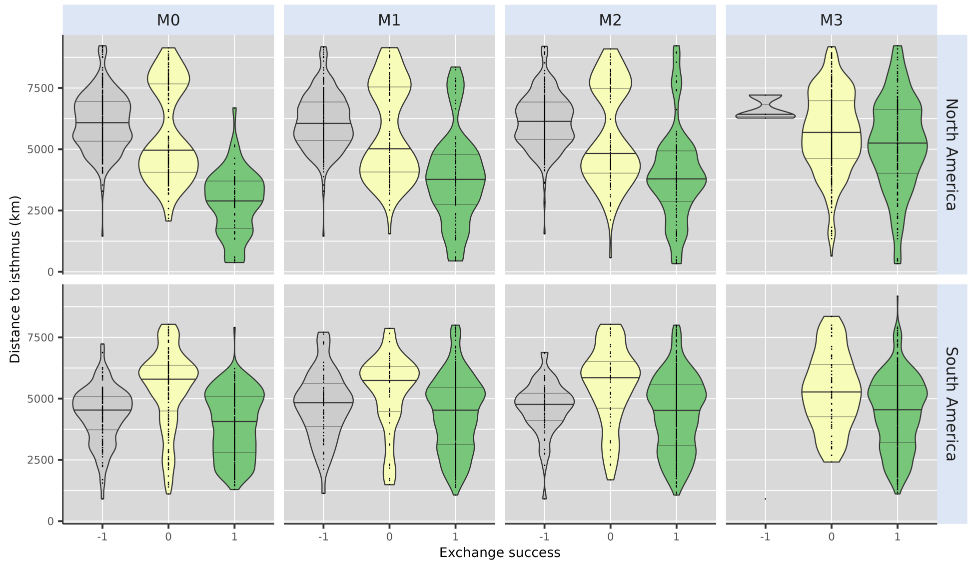

Figure S 8: Comparison of the initial distance to the isthmus between successful and unsuccessful simulations of each of our four models including simulations that crashed (*i.e.*, resulted in no living representative at the final time step), either starting from North (top) or South (bottom) America. Successful simulations ($Exchange\ Sucess = 1$) are displayed in green, unsuccessful simulations ($Exchange\ Sucess = 0$) in yellow and simulation that crashed ($Exchange\ Sucess = -1$) in grey. Dashed lines on top and bottom of the distribution represent the 25% and 75% quantiles whereas the thicker full line in the middle shows the median (50% quantile).

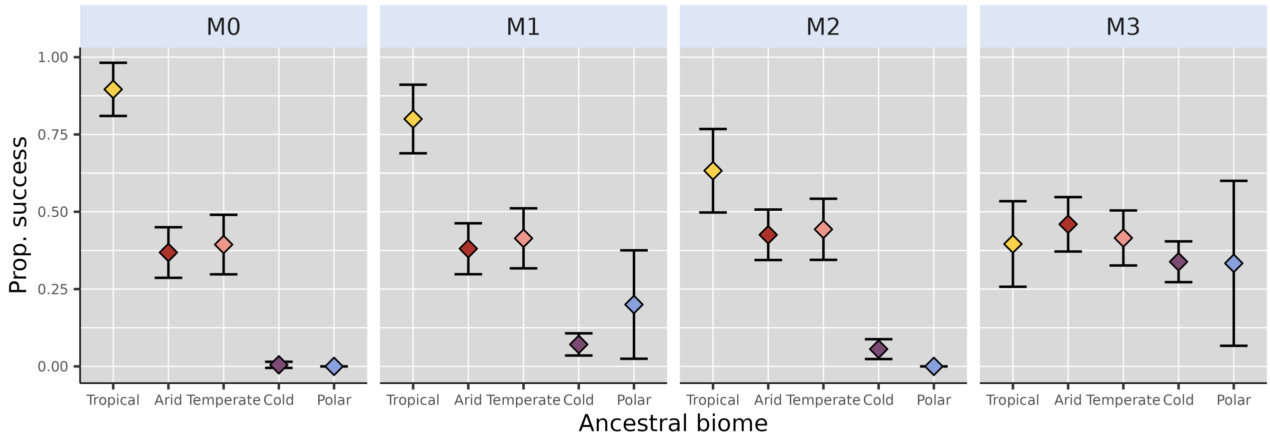

Figure S 9: Proportion of successful simulations starting from North America – after standardising initial distances to the isthmus with simulations starting from South America (see *STAR Methods* and Figure S2) – for each scenario. Coloured dots and error bars respectively show proportion estimates – ratio between number of successful simulations and total number of simulations starting for a given biome – and their associated Binomial-derived 95% confidence intervals.

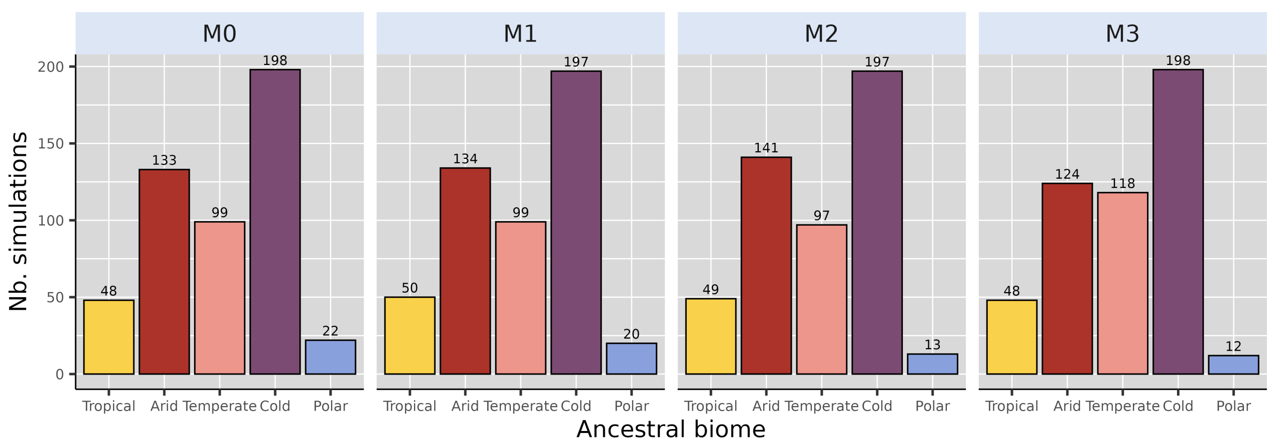

Figure S 10: Biome where the ancestral species of each of the 500 simulation originated, for each modelling scenario, starting from North America after standardising initial distance to isthmus with that of simulations starting from South America (see *STAR Methods* and Figure S2). We removed simulations starting from a cell that could not be assigned a biome from our counts, hence the reason why some are not exactly summing to 500.

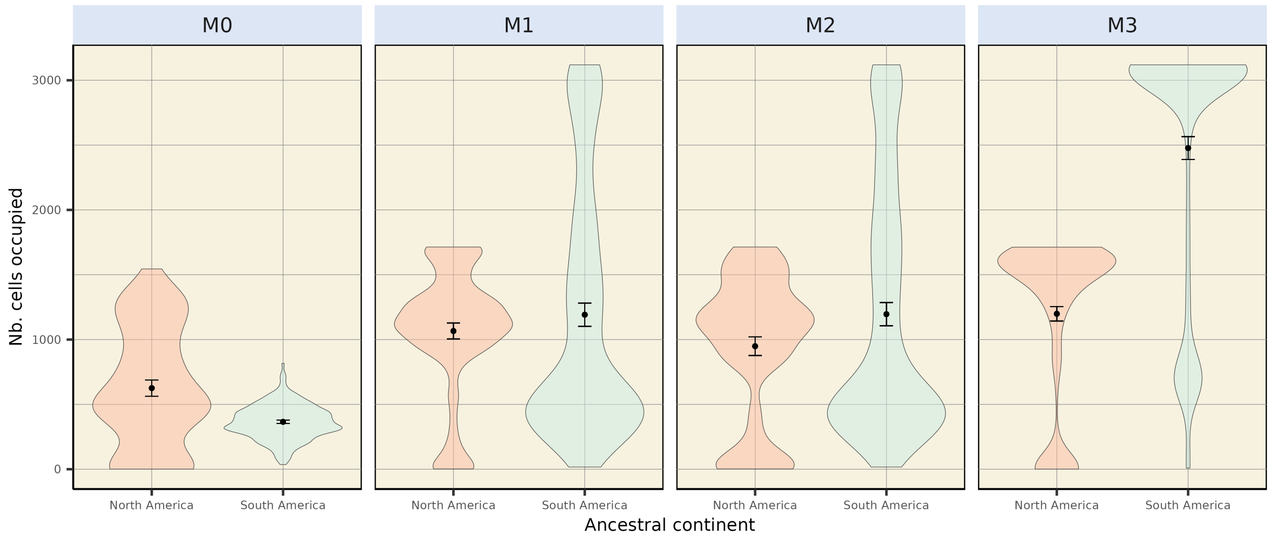

Figure S 11: Absolute colonised area, expressed as the number of foreign continent's occupied cells for each successful simulation generated under each model ($M0$–$M3$).

Supplementary tables

Table S 1: Fixed parameter values for gen3sis simulations.

Name	Value	Description
`divergence_factor`	0.05	How much divergence disconnected populations accumulate across iterations
`max_species`	10000.00	Maximum number of simulated species a simulation can handle (if falls above, the simulation crashes)
`max_coex_species`	1000.00	Cell carrying capacity
`start_species`	1.00	Number of species to start with
`init_ab`	500.00	Initial abundance of the start species
`grid_cell_distance`	111.00	Distance (in km) between two grid cells of the landscape at the equator

Table S 2: Proportion of simulations that crashed – *i.e.*, leading to no living representative at the end of the simulation – for each modelling scenario ($M0-3$) depending on the ancestral area (`Start` column). "North America Standardised" refers to simulations carried out from a North American ancestor after standardising initial distances to Isthmus with simulations starting from South America (see *STAR Methods*)

Start	M0	M1	M2	M3
North America	0.590	0.546	0.554	0.006
North America Standardised	0.470	0.394	0.416	0.000
South America	0.204	0.134	0.118	0.002

Table S 3: References for the silhouettes used in Figure 1, from Phylopic.

Silhouette	**Author**	**License**
Equus (Amerhippus)	Julián Bayona	CC BY-SA 3.0
Smilodon fatalis	Public Domain	CC0 1.0
Cuvieronius hyodon	Henry Fairflield Osborn	CC BY-SA 3.0
Lama guanicoe	Public Domain	CC0 1.0
Toxodon platensis	Julián Bayona	CC BY-SA 3.0
Megatherium americanum	Julián Bayona	CC BY-SA 3.0
Tayassu pecari	Gabriela Palomo Muñoz	CC BY 4.0
Saimiri	Public Domain	CC0 1.0
Dasypus novemcintus	Public Domain	CC0 1.0
Tapirus augustus	Julián Bayona	CC BY-SA 3.0

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Universida de Vigo

Lucas Buffan
Institut des Sciences de l'Évolution de Montpellier

February 4th, 2026

Dear Editor-in-Chief and Editorial Board members of *Ecology Letters*,

Please find enclosed our manuscript entitled "***Palaeoclimate alone fails to explain the asymmetry of the Great American Biotic Interchange***", which we are pleased to submit for consideration as a letter in *Ecology Letters*.

The Great American Biotic Interchange (**GABI**) is one of the most famous biogeographic experiments that shaped modern ecosystems. It characterises the biotic admixture that occurred between the Americas, catalysed by the closure of the Isthmus of Panama in the Mio-Pliocene. A puzzling feature of the GABI is that this exchange has been **asymmetric**: a larger number of North American taxa successfully established to the South than the other way around. Despite being the scope of a wealth of study, the **underlying causes behind such an asymmetry remain unresolved**. This study envisages the asymmetry of the GABI from an innovative perspective, and proposes valuable insights into the long-debated processes that shaped this textbook example of macroecological imbalance.

Coupling high-resolution **palaeoclimatic reconstructions** covering the last 5 million years with cutting-edge spatially-explicit **eco-evolutionary simulations**, we **test** long-standing **hypotheses relating the asymmetry of the GABI to palaeoclimate dynamics**. Contrary to empirically-derived expectations, our simulations indicate that, when climate is the only constraint for species dispersal, **North America is easier to reach from South America than the other way around**. In addition, most of our simulated interchanges involved taxa from **tropical biomes** of both continents, in line with predictions of biome selectivity made from phylogenies, but not from the fossil record. This suggests that classical views of the factors shaping the asymmetry of the GABI drawing on fossils could have been affected by a taphonomic bias. Our computer experiments **challenge the paradigm** solely relating the **asymmetry of the GABI to palaeoclimate dynamics**, suggesting that other factors must have been at play to fashion the disproportionate success of North American taxa compared Southern ones during the GABI.

To our knowledge, our study is the first to provide **mechanistic evidence** to the question of the asymmetry behind the GABI. This work takes full advantage of **simulated data**, a powerful source of evidence unaffected by any upstream bias related to preservation or sampling, to provide a **new perspective to long-standing questions** in the fields of biogeography and palaeontology. It constitutes a **new step towards a better understanding of the processes underlying large-scale ecological events that shaped the biosphere**. In addition, the research approach implemented here is outstandingly new to all co-authors − myself included −, who never published on eco-evolutionary modelling. Hence, we think that our work fits the scopes of *Ecology Letters*, and hope that you will share our excitement for the study presented.

Also, for qualified reviewers, we would like to recommend:

- Dr. Alexander Skeels (alexander.skeels@anu.edu.au)
- Dr. Théo Gaboriau (Theo.Gaboriau@unil.ch)
- Dr. Roniel Freitas-Oliveira (freitasronielbio@gmail.com)
- Dr. Christine D. Bacon (christine.bacon@bioenv.gu.se)
- Dr. Juan D. Carrillo (juan.carrillo@mnhn.fr)
- Prof. Darin A. Croft (dcroft@case.edu)

Best regards,

Lucas Buffan, on behalf of my co-authors

Palaeoclimate alone fails to explain the asymmetry of the Great American Biotic Interchange

Lucas Buffan[1]*, Fabien L. Condamine[1], Sara Varela[2]

[1] Institut des Sciences de l'Evolution de Montpellier, Université de Montpellier, CNRS, IRD, Place Eugène Bataillon, 34095 Montpellier cedex 5, France.

[2] Centro de Investigación Mariña, Departamento de Ecoloxía e Bioloxía Animal, Universidade de Vigo, Campus Lagoas-Marcosende, 36310 Vigo, Spain

* Corresponding author: lucas.l.buffan@gmail.com

Statement of authorship

S.V. and L.B. designed the project. L.B. ran the simulations, performed the analyses, produced the figures and wrote the manuscript with inputs from all co-authors.

Data availability statement

R scripts for generating, analysing and visualising the data are available at https://github.com/Buffan3369/GABI-gen3sis.git. Simulation landscapes and config files are available at https://doi.org/10.6084/m9.figshare.31160704.

Keywords

Biotic Interchange; Eco-evolutionary simulations; Macroecology; Palaeoclimate

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